

NVSHOP.LK – Comprehensive Project Documentation

STATUS: 97%
COMPLETE

Production & Viva Ready

Enterprise Full-Stack E-Commerce Platform for Genuine Branded Mobile Accessories (Anker, UGREEN, Baseus)

2nd Year Software Project

■ **Showroom:** 185/1/2B New Road, Ambalangoda 80300, Sri Lanka | ■ **Hotline:** +94 76 989 0079 | ✉ **Email:** nvshopamba@gmail.com

1. Executive Summary & Completion Status

NVSHOP.LK is a production-grade full-stack e-commerce web platform engineered for an authentic retail shop in Ambalangoda. It features a complete customer journey, real-time inventory management, dual-channel OTP verification (Email + Sri Lankan Mobile SMS), and order tracking with automated invoices.

Module / Domain	Status	Progress	Delivered Capabilities
Catalog & Browsing	Completed	100%	48 authentic products, 11 categories, real-time search, multi-criteria filtering
Product Details & Gallery	Completed	100%	Multi-angle image switcher, Rs. 1000 OFF badge, dual actions (Add to Cart / Buy Now), specs
Dual OTP Authentication	Completed	100%	Notify.lk Sri Lanka SMS Gateway + Gmail SMTP HTML Email OTP verification
Shopping Cart & Promos	Completed	100%	Exact NVSHOP.LK Cart UI, free islandwide delivery progress bar (Rs. 10k threshold), NV10 coupon
Checkout & Orders	Completed	100%	Full Sri Lankan address form, Cash on Delivery support, instant order creation
Order Tracking & Invoices	Completed	100%	Live Status Stepper (PROCESSING → SHIPPED → DELIVERED), printable receipts
Customer Profile & Warranty	Completed	100%	Order list, digital brand warranty certificates with serial numbers & remaining days calculation
Product Comparison	Completed	100%	Side-by-side spec comparison table for up to 4 devices simultaneously
Admin Management Suite	Completed	95%	Product CRUD, stock updates, order status switcher, sales revenue summary metrics
External Integrations	Completed	100%	MySQL Database, Notify.lk REST API, Gmail SMTP (Port 587), Google Maps Showroom Link

2. Full-Stack Technology Architecture

■ Frontend (Client-Side)

- **React 18** (Single Page Application architecture)
- **Vite v8.0** (Lightning fast build tool and HMR)
- **Zustand** (Global state stores for Cart, Auth, Wishlist, Compare)
- **Tailwind CSS & Vanilla CSS** (Responsive modern UI)
- **React Router DOM v6** (Client-side routing with aliases)
- **Lucide React** (High-clarity vector icons)
- **Axios & Fetch API** (REST API communication)

■ Database Infrastructure

- **MySQL (via XAMPP / MariaDB)** on Port 3306
- **Database Name:** gpsd_project
- **Core Relational Entities:** users, products, orders, order_items
- Foreign key constraints and automated schema migration

■ Backend (Server-Side)

- **Spring Boot v4.1.0 / Java 17 LTS**
- **Embedded Apache Tomcat 11** (Port 8080)
- **Spring Security & BCrypt** (Password encryption & token auth)
- **Spring Data JPA & Hibernate ORM 7** (Data persistence)
- **HikariCP** (Connection pool with auto-reconnect)
- **Jakarta Validation API** (@Valid, @NotBlank, @Email)
- **MimeMessage JavaMailSender** (HTML email delivery)

■ Live Third-Party Cloud Integrations

- **Notify.lk SMS REST API:** Live Sri Lankan SMS OTP delivery to Dialog, Mobitel, Airtel, and Hutch (User ID: 32620)
- **Google Gmail SMTP (Port 587 TLS):** HTML verification & invoices
- **Google Maps:** Direct showroom coordinates linking

3. 6-Member Group Project Viva Allocation Matrix

To ensure full marks in the viva evaluation, all SRS requirements and project code are systematically distributed among 6 members:

Member	Assigned Domain & Responsibilities	Key Code Files & Technical Concepts to Explain
Member 1	Architecture & Dual OTP Security Backend lead	• Spring Boot REST Controller architecture (AuthController.java) • BCrypt Password encryption & Security filter chain • Notify.lk SMS REST API live HTTP payload dispatch & SMS verification (SmsService.java) • Gmail SMTP TLS port 587 email verification dispatch (MailService.java)
Member 2	Catalog, Search & Comparison Catalog specialist	• ProductController.java and ProductService.java data flow • Spring Data JPA Repository query methods with price & category criteria • Client-side dynamic search filtering and sorting algorithms in React (Products.jsx) • 4-device side-by-side spec comparison matrix (Comparison.jsx)
Member 3	UI/UX, Product Specs & Warranty Frontend UI designer	• Responsive layout system using Tailwind CSS and CSS Grid • Product Details showcase: Rs. 1000 OFF badge, interactive gallery thumbnails (ProductDetails.jsx) • Zustand Wishlist store integration with local persistence (Wishlist.jsx) • Digital warranty certificate engine with serial numbers & remaining days (UserProfile.jsx)
Member 4	Cart, Promo Engine & Checkout Commerce workflow	• Zustand Cart State management (useCartStore) for quantity & line totals • Free Islandwide Delivery progress calculation (Rs. 10,000 threshold bar in Cart.jsx) • Dynamic coupon validation engine (NV10 / WELCOME1000) • Sri Lankan checkout form validation with Cash on Delivery selection (Checkout.jsx)
Member 5	Order Pipeline & Live Tracking Order automation	• Order Entity One-to-Many mapping (Order ↔ OrderItem) • Automated transactional order processing (OrderService.java) • Real-time delivery status visual stepper (PROCESSING → SHIPPED → DELIVERED in Orders.jsx) • Automated MimeMessage HTML invoice generation sent to customer email
Member 6	Admin Portal & DB Infrastructure DB & Admin lead	• MySQL schema design and HikariCP connection pooling configurations (application.properties) • Admin protected route authorization and credential validation • Full Product Inventory CRUD (Create, Read, Update, Delete in Admin.jsx) • Live Order status switcher and sales revenue summary metrics

4. Live Demo Endpoints & Credentials

Role / Service	URL / Port	Credentials & Instructions
Customer Web Portal	http://localhost:5173	Sign up with any +94 phone or email, or click ' Instant Preview with Demo Customer '
Admin Management Portal	http://localhost:5173/admin	Email: admin@example.com Password: admin123
Backend REST API	http://localhost:8080/api	Spring Boot 4.1 REST API (Products, Auth, Orders, Categories)
MySQL Database	localhost:3306	Database: gpsd_project User: root Pass: (blank)
Notify.lk SMS Gateway	app.notify.lk/api/v1/send	User ID: 32620 Sender ID: NotifyDEMO